[SOIL CLASSIFICATION IN THE FIELD]

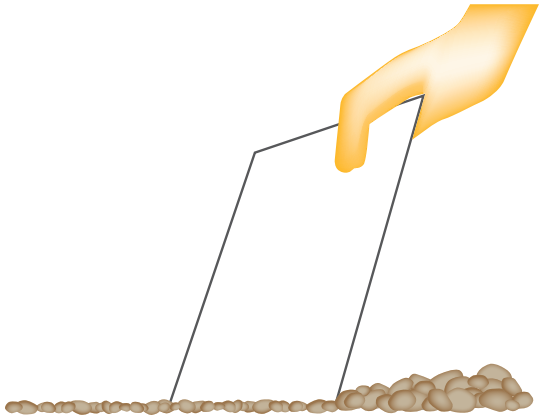
Classification systems require laboratory measurements such as the sieve analysis or the Plasticity Index test. However, some simple field tests can be used to classify various soils when complete laboratory facilities are not available. The tests are used to determine gradation, plasticity and dispersion.

Gradation/Particle Distribution – To test the gradation of dry soil, spread a sample of the soil on a flat surface. Use a piece of stiff paper or cardboard as a rake to sort the larger soil particles to one side. Estimate the percentage of particles larger than 5 mm (3/16") and the percentage of fines (too small for the individual grains to be seen by the unaided eye). Also, estimate whether the larger particles are uniform in size (poorly graded) or have large, medium and small sizes (well graded).

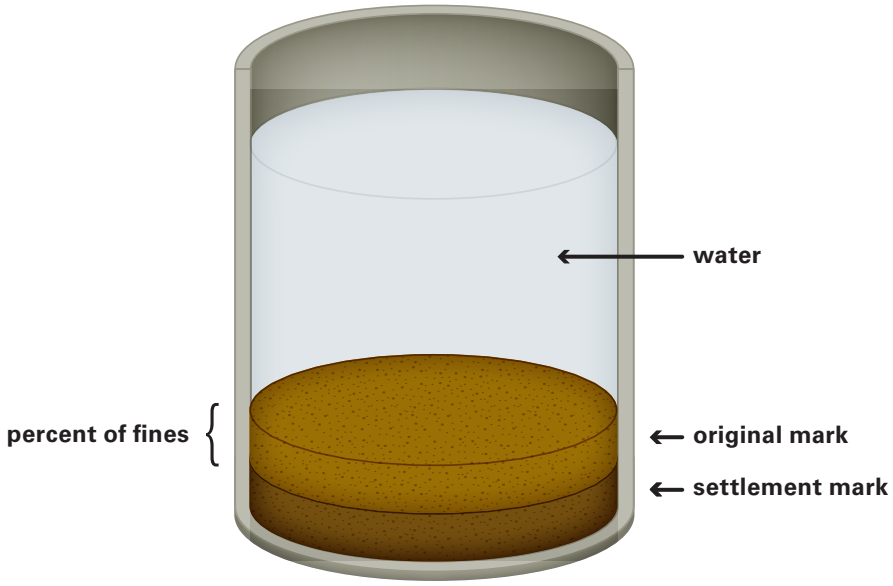
If the soil is wet, break a lump apart with a pencil and make percentage estimates in the dry soil method. To find the percentage of fines, fill a clear glass with 3 mm (1/8") of water. Then, add enough soil to fill the glass 1/4 full. Add water until the soil is just covered.

Mark this level with a rubber band. Fill the jar 3/4 full with water and stir the mixture vigorously. Let it settle about a minute and a half and mark the height of soil that has settled out. The difference between the two marks represents the percentage of fines.

GRADATION TEST

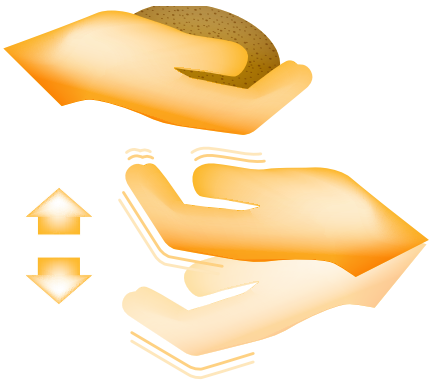


PERCENT OF FINES



Plasticity of Fine Grained Soils – You can perform one or more field tests to estimate the plasticity of a soil.

SHAKING TEST

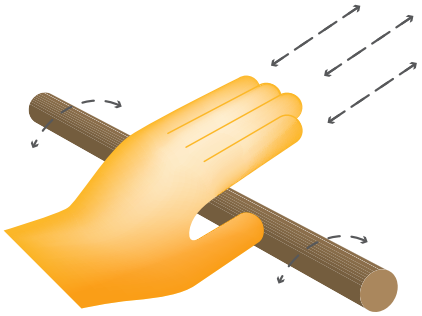


- **Shaking Test –** Pick up a lump of fine-grained soil and knead it together, working out as many large grained particles as possible. Add water gradually and knead the soil until it begins to get sticky. Hold the ball of soil in the palm of one hand and tap the back of that hand with the fingers of the other hand.

  If the ball gets shiny and wet on the surface, it is mostly fine sand or silt. Clays have little or no reaction to this test and simply get messy.

- **Toughness Test –** Take about half the ball of soil and knead it between the thumb and forefingers to dry it out. Then, attempt to roll the soil sample into a thread or "worm" If a worm cannot be formed at all, the soil is definitely a silt or fine sand. Highly plastic soils take a long time to dry out. They get hard and waxy, and considerable pressure is required to form a worm that just breaks at a diameter of 3 mm (1/8 in).

TOUGHNESS TEST



- **Dry Strength Test –** Take the other half of the ball of soil and knead it into a ball. Set it aside to air dry. When the soil is dry, crush it and select a jagged, pointy fragment. Try to crush this fragment between the thumb and forefinger. Silt will turn to powder with little effort. Clay will be like a rock and almost impossible to crush with the fingers.

- **Hand Washing –** After handling silts and sands, the fingers will feel dusty and rubbing the fingers together will almost clean them. Water flowing gently from a faucet will rinse off the soil. When clays are handled, a crust will form on the fingers that cannot be rubbed off when dry. Water will not rinse it off. The hands must be rubbed together under water to cleanse them.

HAND TEST



- **Hand Test –** Pick up a handful of soil. Squeeze it; then open your hand. If the soil is powdery and will not retain the shape made by your hand, it is too dry. If it shatters when dropped, it is too dry. If the soil is moldable and breaks into only a few pieces when dropped, it has the correct amount of moisture for proper compaction. If the soil is plastic in your hand, leaves traces of moisture on your fingers and stays in one piece when dropped, it has too much moisture for compaction.

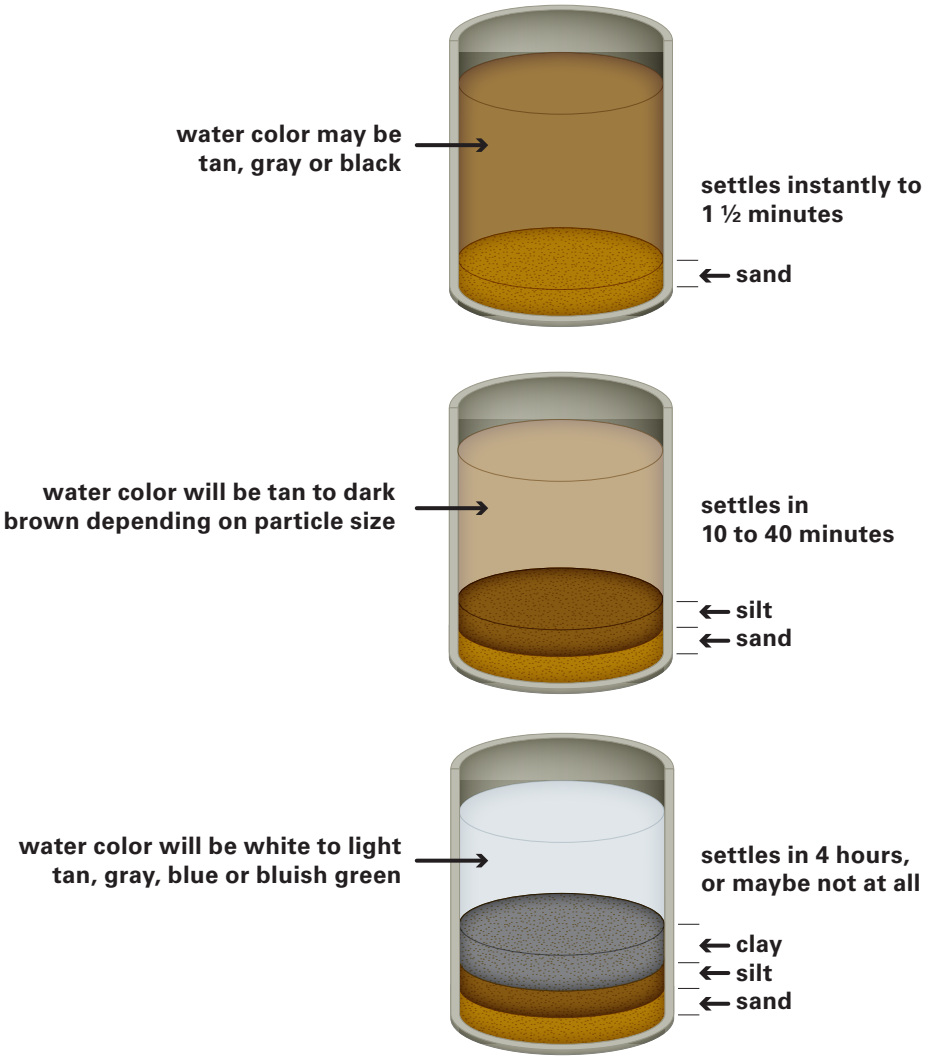
- **Dispersion Test –** In addition to the field tests just described, the Dispersion Test can be used to determine percentages of soil grain sizes as well as an indication of how difficult it will be to compact the soil. All that is needed is a clear glass, water and a representative soil sample.

  Fill the glass 1/4 – 1/3 full with the material. Then, fill the container with water to within 15 mm (1/2 in) of the top. Stir the mixture well and observe how the material settles.

  The material will settle in three distinct layers. The sand at the bottom, silt next and finally, clay. Besides showing the various groups, the results will show whether the soil is well or poorly graded. Although the silt and clay particles are smaller than the eye can see, gradation changes can be observed by color differences. Also, the longer it takes a layer to settle, the smaller the particles.

  There are several things that can be learned from the dispersion test. It will show the basic materials and gradation of each, and the settling time will indicate the fineness of the particles. In most cases, a single particle size (poor gradation) and a small particle size will indicate a less stable construction material than a mix where there is a good gradation of all particle sizes. These materials are difficult to compact because the grains continue to shift under the machine.

DISPERSION TEST



[SUMMARY OF FIELD TESTS]

Soil Types	Description
Clays	No reaction to the shaking test; a rough worm that dries out slowly; a crusty dry residue that is hard to remove from the hands.
Silts	Rapid reaction to the shaking test; a weak or crumbly worm; powdery residue that is easily wiped off or washed off the hands.
Silt and clay mixtures	Intermediate or conflicting reactions to hand tests.
Sand or gravel with fine clays	Enough clay to soil the hand if a wet sample is kneaded, but not enough to allow a lump of clay to be formed.
Sand or gravel with silt fines	Any mixture with dusty or fairly gritty fines.
Clean sands and gravels	Water added to these soils sinks immediately without making any mud.
Shot or ripped rock	Jagged material not having enough smaller material to fill the voids.




CAT
CAT



Unit 3
PHYSICS OF COMPACTION

Understanding the physics of compacting different soil types and the effect of different machines and their compactive capabilities is the key to meeting specified soil density in the most cost-efficient manner.

[FACTORS THAT INFLUENCE VIBRATORY COMPACTION]

Vibratory compaction of soil is a complex process. Many different factors influence the overall compaction effort. All of these factors that influence compaction must be considered in total, not independently. It is the combined characteristics of the compactor and of the soil it is acting upon that determine the degree of compaction effort. And, project specifications determine if the compaction effort is adequate. The factors or characteristics influencing vibratory compaction can be divided into three categories:

1. Material and jobsite related characteristics
2. Project specification related characteristics
3. Machine related characteristics

Material and Jobsite Related Characteristics

- **Soil Type –** A given soil type will have unique compaction characteristics; soils that are more difficult to compact call for heavier compactors.
- **Gradation –** The gradation of a material is the range of particle sizes present. Ideally, there are relatively equal amounts of all grain sizes with no predominant size present.
- **Uniformity –** A soil material is a mix of many soil types and particle sizes. Uniformity can be thought of as the degree to which all of the composite materials are well mixed and dispersed evenly throughout the soil. A uniform soil mix is homogenous and will compact consistently; a soil that lacks uniformity will exhibit inconsistent compaction.

  The Uniformity Coefficient (C_u) in soil mechanics is a parameter to describe the particle size distribution (grading curve) of a soil. It provides information on how uniformly the grain sizes of a soil are distributed. In DIN EN ISO 14688-2:2004, C_u is defined as the ratio of the diameter d_{60} during sieving of 60 percent to the diameter d_{10} at 10 percent passing. The ratio represents the slope of the range of the grain size curve between 10 percent and 60 percent of passes (through the sieve).

CALCULATING THE UNIFORMITY COEFFICIENT

$$C_u = \frac{d_{60}}{d_{10}}$$

The value of C_u allows the following statements on the soil:

C_u <5 = uniform soil
C_u 5-15 = non-uniform soil
C_u >15 = extremely non-uniform soil



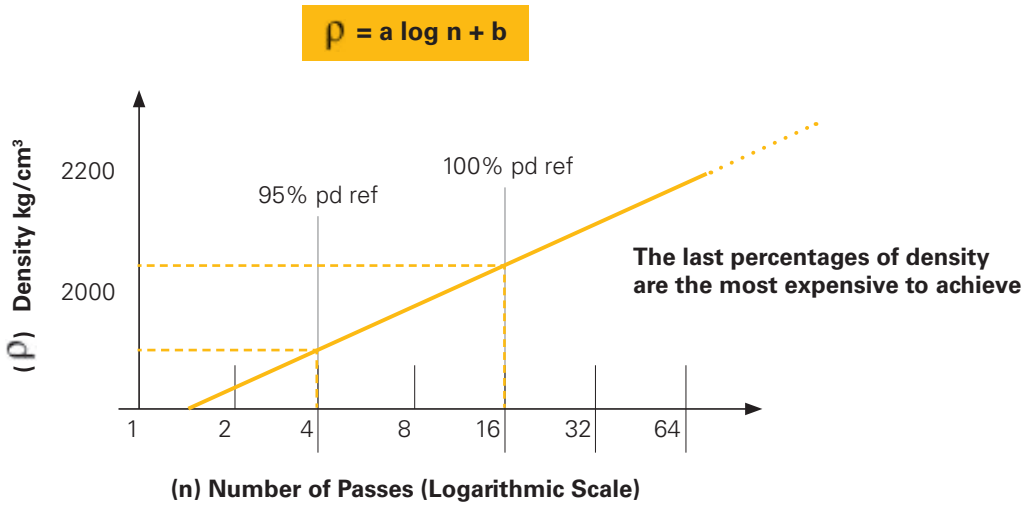




- **Texture –** Individual soil types possess different surface textures, which have an effect on the compaction characteristics of the material. Soil types with coarse texture create high friction between the particles, requiring more energy from the compactor to loosen their bonds, thereby allowing them to reposition in a denser state. Smooth textured particles slide more easily over one another, requiring less effort to compact.
- **Grain Shape –** Like texture, the shape of the particles can also affect compaction of a soil. Jagged shapes tend to have greater friction bonds that require greater effort to compact. Smooth, rounded shapes slide more easily with less compactive effort.
- **Initial Density –** Materials with higher initial density will require less compactive energy than those with lower density. This will affect productivity, as less dense material may require more passes.
- **Moisture Content –** Moisture is the single most important factor to consider when compacting soil. Too little moisture and particles will not adhere to each other. Too much moisture causes particles to displace easily. Every soil type has a moisture content that is ideal for optimal compaction (defined by Proctor Test).
- **Aggregate Strength Characteristics –** Every soil type has a different compressive strength based upon how the aggregate was formed.
- **Subsoil Base and Its Supporting Capability –** A structure is only as strong and resilient as the foundation that supports it. If the subsoil lacks the capability to support a road, it is likely that compaction of the sub-base and base layers will be difficult. Unsuitable soils can be improved by chemical or mechanical stabilization.

THE COST OF DENSITY

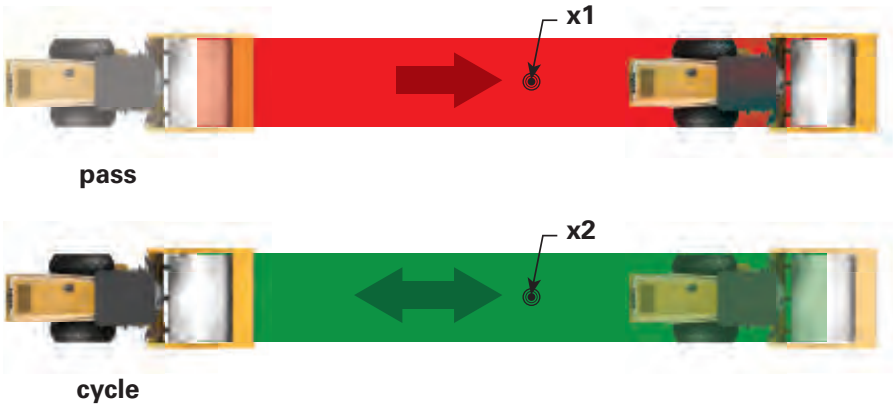
With increasing number of passes, density increases



Project Specification Related Characteristics

- **Compaction Target –** Typically, this is established by the administration of the Standard Proctor or Modified Proctor test and is specified as a certain percentage of the maximum dry weight density resulting from that test—in the example, 95 percent of Standard Proctor. Generally, the higher the compaction target, the more passes are required. It is always the most difficult to achieve the final percentage points of compaction.
- **Lift/Layer Thickness –** When using a compactor of a given size, the layer thickness will influence productivity. A thick layer will require more passes than a thin layer.

COUNTING PASSES



- **Number of Passes –** This refers to the number of times a compactor will pass over an area of ground. Caterpillar defines a pass as a single trip over an area in either forward or reverse direction and a cycle as two consecutive passes over an area, typically one in the forward direction and one in the reverse.

  Knowing the pass count can be important, particularly when an area is identified that is not meeting compaction specifications. If a jobsite receives the same coverage (number of passes), and one area fails but the rest pass, it can help you narrow the number of causes.

In cases where a method specification is used, the number of passes (with an appropriately sized compactor on a lift of specified composition and thickness) will be specified. In those cases, engineers have determined from past experience that this number of passes will be sufficient to meet the compaction target.

Machine Related Characteristics

The design of the machine is important to the dynamics of compaction. Influential factors include: frame size, overall weight, wheelbase, ratio of machine weight supported over the front drum to tires, and balance of machine weight from left to right of the machine. The list continues with factors such as drum diameter, drum length, drum mass, shock isolators, eccentric weight mass, and the distance between the eccentric weight center of gravity and drum axle. Even the weight of fuel and the operator influences the performance of the compactor. The manufacturer carefully considers all these factors when designing each machine.



Vibratory compaction involves a drum (static weight) that is moving up and down (amplitude) very rapidly (frequency) and moving forward (working speed) over non-homogeneous material.

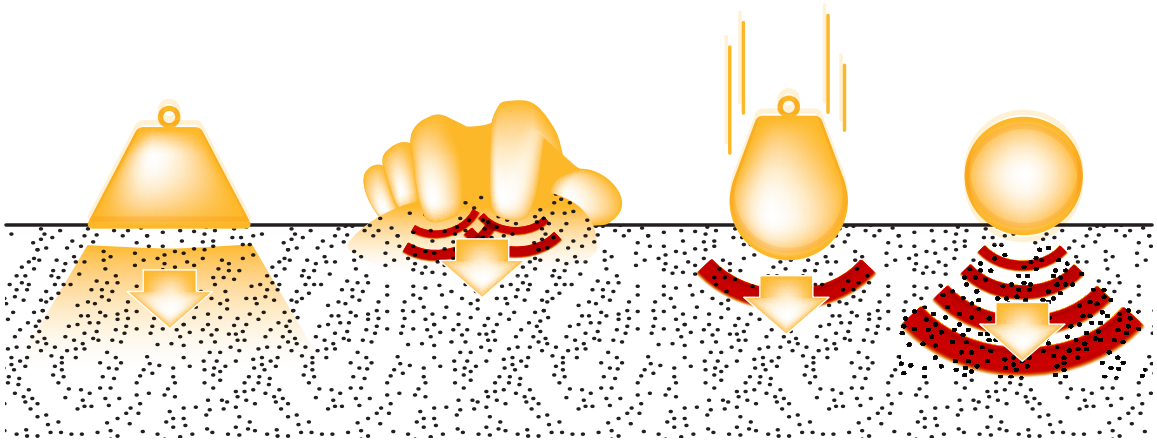
Of course, frequency, amplitude and working speed are variables controlled by the operator. They will be discussed later in Unit 3.

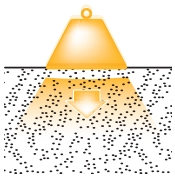
All these variables mean that it's not always easy to set up a compactor on a given job to achieve ideal compaction results. The objective in vibratory compaction is to find a point of maximum transmitted force into the material to be compacted. This occurs when the sum of all the factors—material characteristics, compactor characteristics, amplitude, frequency and speed—is contributing optimally to the compactive effort required for meeting the project specifications.

[FACTORS THAT INFLUENCE VIBRATORY COMPACTION]

Compaction is the process of compressing a material from a certain volume into a smaller volume. This is accomplished by exerting force and movement over a contact area, causing particles within the material to break their natural bonds with each other and move closer together. The voids between the particles—air, water or a combination of both—are expelled by the combination of force and movement. Four forces are used in compaction:

1. **Static pressure**
2. **Manipulation**
3. **Impact**
4. **Vibration.**

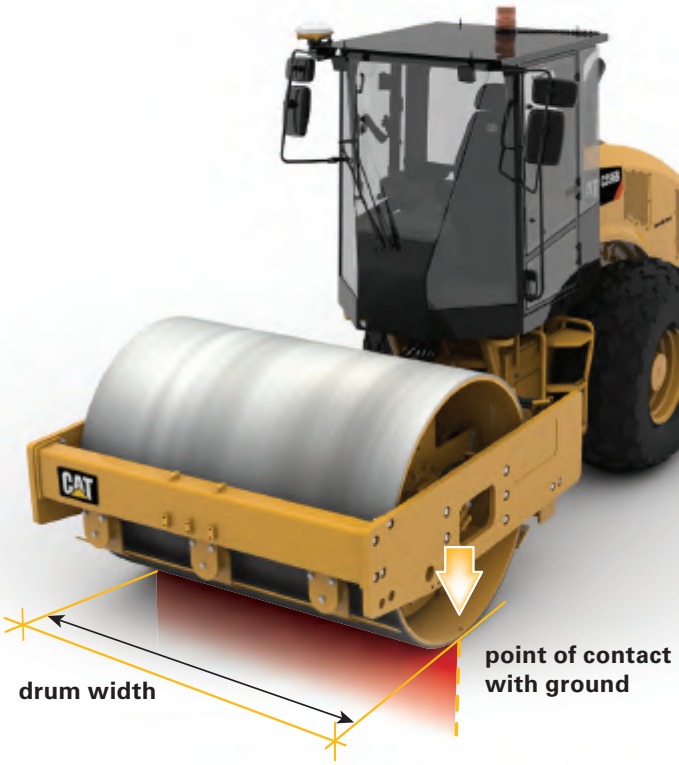




Static Pressure – In static compaction, pressure from the weight of the compactor produces shear stresses in the soil that cause the individual particles to slide across each other. Compaction occurs when the applied force causes individual particles to break their natural bonds to one another and reorient to a more stable position. This compaction force has a greater effect on surface and shallow depth materials. It has a minimal effect on deeper soils.

Static linear load is the measure used by the industry to compare the compaction potential of static, smooth drum compactors. It is the vertical force directly below the width of the drum that creates the shear stresses for compaction. It is calculated by dividing the weight at the drum (axle load) by the drum width. Static linear load is expressed as kilograms per linear centimeter (kg/cm) or pounds per linear inch (lb/in). Compactors with a higher linear load have greater compaction potential and depth of influence.

STATIC LINEAR LOAD

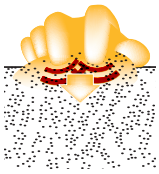


Static Linear Load	
Vibratory Soil Compactors	
5 - 8 Ton	**16 - 22 kg/cm (90 - 120 lb/in)**
8 - 12 Ton	**20 - 30 kg/cm (100 - 200 lb/in)**
12 - 15 Ton	**30 - 45 kg/cm (180 - 250 lb/in)**
>15 Ton	**45 kg/cm+ (250 lb/in+)**
Pneumatic Tire Compactors	**1000 - 3200 kg/wheel (2200 - 7000 lb/wheel)**

For static padfoot drum compactors, tamper foot compactors and sheepsfoot compactors, the amount of tip pressure exerted constantly varies as the number and surface area of the tips that are in contact with the ground changes. The depth of penetration can also affect the calculation. The pressure of the pad faces is expressed in kilograms per square centimeter (pounds per square inch).

Static compaction is used in applications where a gentle touch is required, either due to nearby buildings, materials that are fragile or where surfaces have low load-bearing strength. It is also used in situations where too much compaction force could draw free water to the surface.





Manipulation – Manipulation is a compactive force that rearranges particles into a more dense mass by a kneading process. The process is especially effective at the surface of the material lift. The longitudinal and transverse kneading action are essential when compacting heavily stratified soils such as clay-type soils. Sheepsfoot compactors and staggered wheel, pneumatic tire compactors are specifically designed to deliver this type of compactive force.



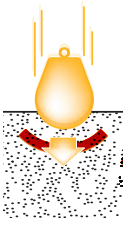
The manipulation force generated by pneumatic tire compactors is a product of two factors: the contact pressure and the wheel load. Adjusting either factor will change the performance of the compactor.

Manipulation compaction is useful to build a well-sealed surface to help the material resist the effects of water and weather.

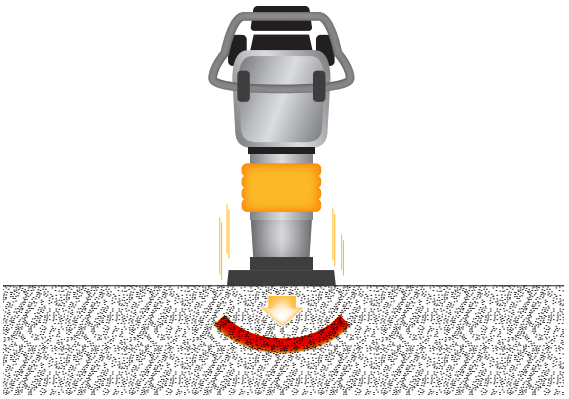
CONTACT PRESSURE FORMULA

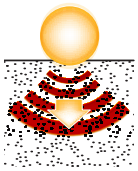
$$\text{CONTACT PRESSURE (kg/cm}^2\text{ : bar)} = \frac{\text{Wheel load (kg)}}{\text{Tire contact area (cm}^2\text{)}}$$

$$\text{WHEEL LOAD (kg)} = \frac{\text{Compactor's operating weight}}{\text{Number of wheels}}$$



Impact – Impact creates greater compaction force than the gravitational force of a static load. This is because a mass in motion has velocity, which is converted to energy at the instant of impact. Impact creates a pressure wave that goes into the ground from the surface. Impacts are usually a series of blows. Impact blows of 50-600 blows per minute are considered low frequency ranges and are used on impact hammers and hand tampers. Impact blows of 1,400-3,000 blows per minute are high frequency and are used on vibratory compactors.





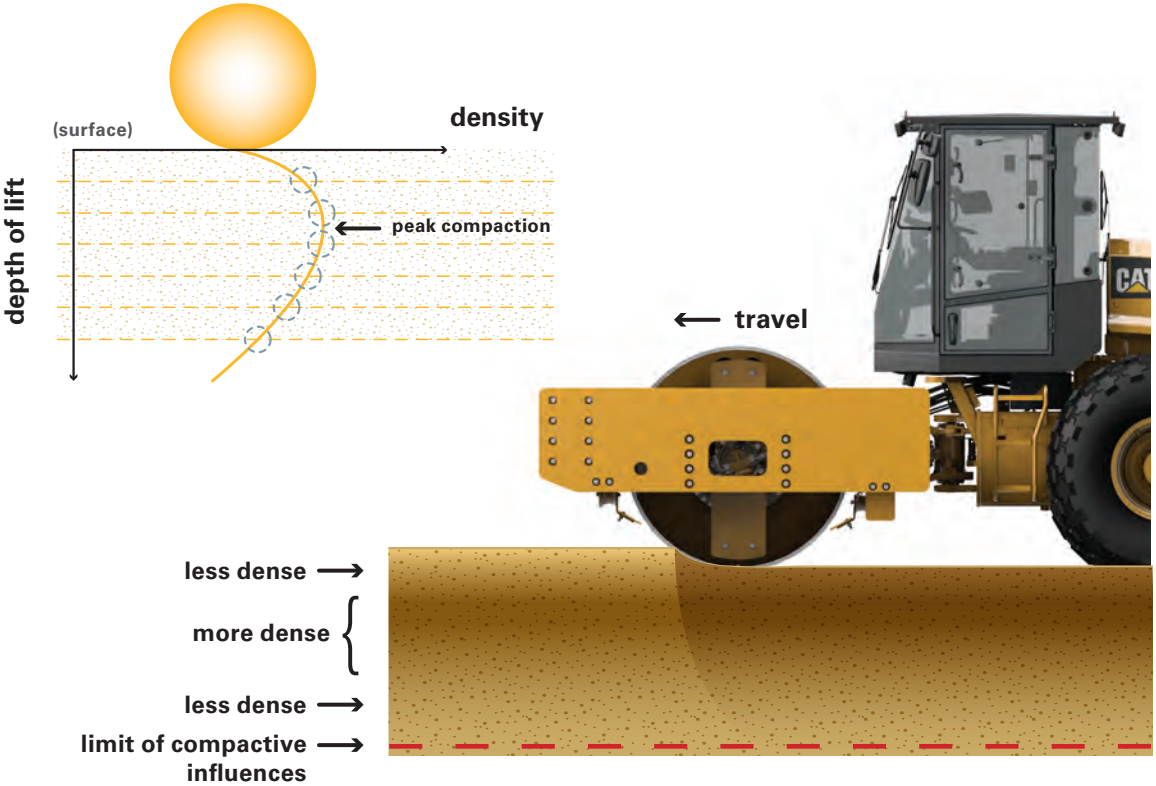
Vibration – Vibration is perhaps the most complex and cost-effective compactive force. More than 90 percent of the compactors sold in today's market are vibratory. This is because vibratory compactors can produce at the same rate as a static compactor that has about three times more mass. Vibratory energy makes a vibratory compactor more efficient than a comparably sized static compactor.

Vibratory compactors produce a rapid succession of pressure waves that spread in all directions. The vibratory pressure waves overcome the shear resistance between the particles of the material being compacted. When pressure is applied, the particles tend to reorient themselves in a more dense (fewer voids) state. To understand how vibratory compactors work, it is necessary to understand the dynamics of vibratory compaction: amplitude and frequency, as well as static linear load and the ratio of vibrating mass to suspended mass.

As a compactor works, the volume of soil compacted will not be uniformly compacted from top to bottom. A compactor of a certain mass will compact soil to a certain depth, but the degree of soil compaction will vary from the surface to the maximum depth of compactive influence. Generally, the surface soil will be less compact, the middling soils will contain a peak compaction, and the maximum depth soils will again become less compact.

Changing operating parameters can influence the depth of the compaction zones and alter the depth of the peak compaction zone, but it still will not change the fact that the soil will vary in compaction from top to bottom. This phenomenon is called the "gradient of compaction," and this data helps describe the suitability of a certain sized compactor or operating setting for specific compaction applications.

GRADIENT OF COMPACTION



It seems logical that when using a rolling compactor, surface soil should be the most dense– however, the peak density actually occurs below the surface, and decreases as depth lessens the influence of the compactor. This is called the Gradient of Compaction.

[DYNAMICS OF VIBRATORY COMPACTION]

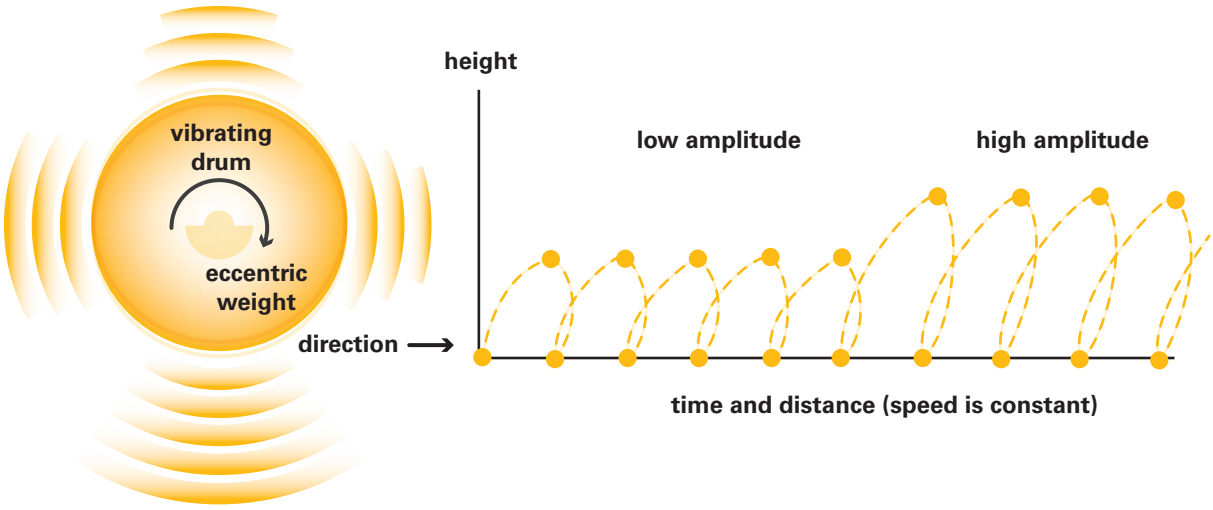
Amplitude – Amplitude is the measure of the distance of vertical movement from rest position to peak position of a vibrating drum. Manufacturers promote this nominal value, which is measured on a suspended drum. The real, working amplitude, however, is the product of the nominal amplitude and the coefficient of amplification, which is a ratio of the transmitted frequency to the resonant frequency of the machine and the ground being compacted. It can be more useful to think of amplitude as the distance the drum travels into the ground as it displaces and compacts soil.

By modifying the amplitude, an operator can vary the force and the movement (acceleration) of the drum on the material.

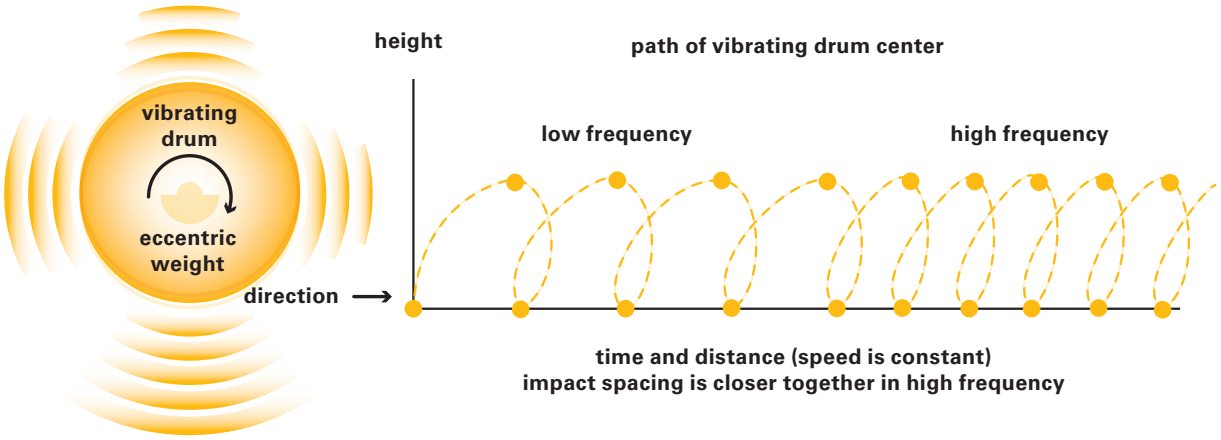
As soil nears maximum density, there comes a point where the soil can't absorb the compactive energy the vibratory compactor is delivering. At this point, the drum can bounce off the surface and a vibratory cycle can occur while the drum is suspended in the air. This phenomenon is called "de-coupling," or "double-jumping," and is accompanied by a distinct and uncharacteristically vigorous vibration affecting the entire machine. De-coupling can damage the machine and produce undesirable results on the soil being compacted, such as decompaction.

To stop de-coupling, the operator needs to decrease the amount of energy the machine is delivering to the soil by simply reducing the amplitude, which will lessen the amount of compactive force applied to the soil. Alternatively, the operator could work in static mode.

AMPLITUDE



FREQUENCY



Frequency and Speed – Frequency is a measure of the number of complete cycles or revolutions of the eccentric weights around the axis of rotation over a given length of time. Frequency is usually expressed in units of hertz (Hz) or vibrations per minute (vpm). Typically, frequencies between 23-35 Hz (1380-2100 vpm) are used depending on material and amplitude setting.

The relationship between frequency and working speed is sometimes simplified to a rule of thumb that states that frequency and working speed should be adjusted to yield approximately one impact per 25-30 mm (1-1.2 in). A working speed that is too fast can cause "washboarding" (impacts spaced too far apart), and a working speed that is too slow negatively impacts machine productivity. There is an optimum speed and frequency for each compaction application, but they may not yield one impact per 25 mm (1 in). Maintaining uniform compaction is critical, and utilizing automatic speed control features to ensure balance of speed and frequency can help deliver that consistency.

Ratio of Vibrating Mass Over Suspended Mass – One might assume that if a vibratory compactor of a certain mass and amplitude can compact a certain soil with a certain degree of efficiency, then simply applying more mass and more amplitude would make the compactor more efficient on the same soil. This is not necessarily true.

On a vibratory compactor, the vibrating mass (drum) is isolated from the suspended mass (the rear frame), and the ratio between the two is a critical factor in determining the mass and amplitude the compactor can possess. The ratio is carefully balanced to allow the machine to optimize the amount of energy it can safely impart to the soil.



total operating mass



suspended (non-vibrating) mass



vibrating mass



Resonance working against compactive effort

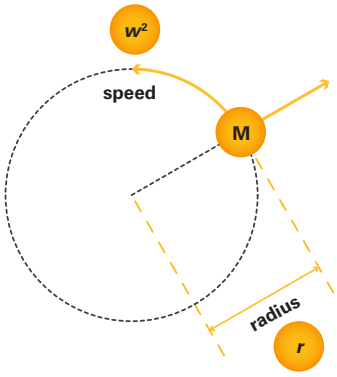


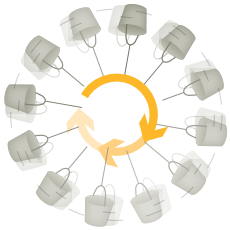
Resonance working together is harmonic convergence

Resonance – When the frequency of imposed vibrations on an object equals the object's natural frequency, the object will vibrate at resonance. In vibratory compaction, resonance is very important.

The interaction between the material being compacted and the vibratory machine causes the material and the machine to vibrate. The eccentric weights rotating inside the drum maintain this vibration at a frequency equal to the rpm of the eccentric weight shaft. At some conditions of the given frequency, the machine and material vibrate at resonance. The conditions that produce resonance depend not only on machine characteristics, but also upon the nature of the material to be compacted and its achieved degree of compaction. The ideal frequency to achieve the most efficient transmission of compaction energy is about 15 percent higher than the resonant frequency.

CENTRIFUGAL FORCE







centrifugal force = Mw^2r

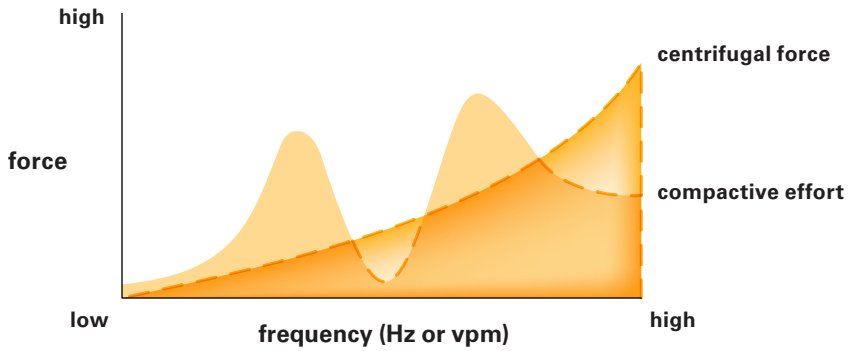
To calculate centrifugal force, the mass (M) of the eccentric weight is multiplied by the radius (r) of the eccentric weight rotation and by the speed of rotation (frequency) squared (w^2). The most significant factor in this equation is frequency.

Centrifugal Force – Vibratory compactors create centrifugal force with an eccentric weight or weights rotating inside a drum. The centrifugal force generated by the drum is analogous to the tug felt when swinging a bucket filled with water. The mass of the weights, their offset distance from the center of rotation to center of gravity, and the speed of rotation all contribute to the production of this force. Centrifugal force is a theoretical calculation and is frequently used to rate the productivity of vibratory soil compactors. But theoretical centrifugal force is not an accurate way to judge the capability of a machine. The true vibrating force depends on a complete interaction between the material being compacted and the machine.

FORCE AND FREQUENCY

This graph shows how theoretical centrifugal force increases as the frequency increases. However, the compactive effort that is actually transmitted to the soil varies as frequency increases. The compactive effort will show multiple "peaks" and "valleys."

Typically, there is a first peak representing a maximum value of the compactive effort, which quickly drops off and is followed by a second peak. Generally, this second peak will produce the higher value, and represents when machine productivity is at its best.



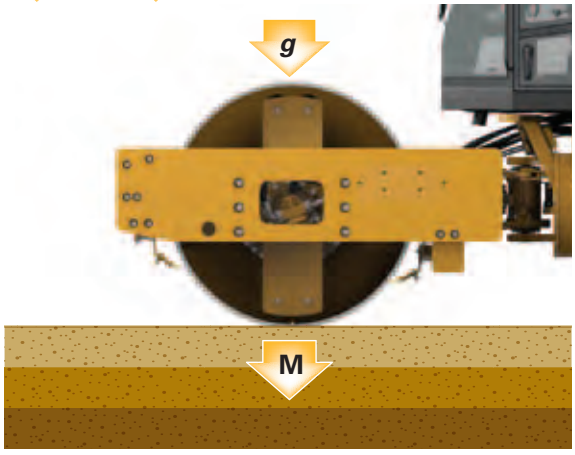
TOTAL APPLIED FORCE (F_{TA})

$$F_{TA} = F_C + F_S$$

Where: F_C is centrifugal force = $1100 \left(M\right)\left(r\right)\left(\frac{N}{1000}\right)^2$

and: F_S is static drum applied load = $M \times g$

- M — *mass of eccentric weight (kg)*
- r — *moment of eccentricity (m)*
- N — *rpm*
- M — *static drum applied mass (kg)*
- g — *acceleration of gravity* $\left(\frac{\text{meters}}{\text{second}^2}\right)$



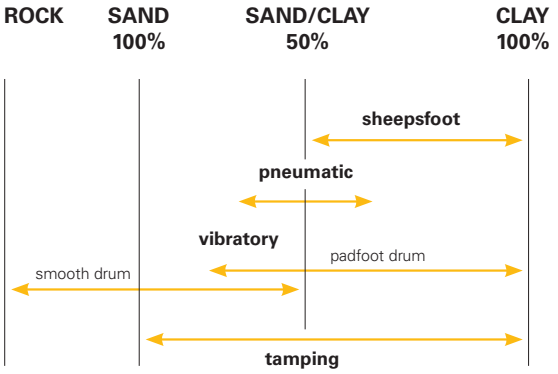
Total Applied Force – Total applied force is regarded as the maximum amount of vibratory energy that a compactor can apply to the ground. It is calculated by adding the static mass of the compactor to the centrifugal force. As with centrifugal force, caution should be used when using this calculation to compare the compactive capabilities of compactors.

[SOIL COMPACTION EQUIPMENT]

Many factors influence the choice of compaction equipment. The type of equipment selected for a project is sometimes based on the contractor's previous experience, by the type of soil, method specification or available equipment. Another consideration is how well a machine will conform to the hauling and spreading operation. Climatic and traction conditions are also important. Sometimes, the contractor's interest in standardizing the equipment fleet plays a role in the decision-making process.

The application chart provides guidelines for matching equipment to job variables and soil types. There is no single compactor that will do all things in all applications. Each type has a definite material and operating range on which it is most economical. In many cases, there are applications where machines of different sizes and types can achieve the compaction target; but choosing the machine that is most suitable will complete the work most economically and efficiently due to reduced passes, reduced fuel use and less working time.

APPLICATION CHART



Vibratory Compactors – Vibratory compactors work on the principle of particle rearrangement to decrease voids and increase density and load bearing strength. They come in two types: smooth drum and padfoot drum. For increased versatility, smooth drum compactors can be equipped with optional padfoot shell kits, which allows the use of smooth drum rollers in padfoot applications, albeit with limited performance.

Smooth drum vibratory compactors generate three compactive forces: static pressure, impact and vibration. Padfoot drum machines generate the same forces, plus they also generate manipulative force. Vibratory compactors provide uniform compaction throughout the lift.

Density is achieved from the forces generated by the vibrating drum hitting the ground. Compaction results are a function of the frequency and amplitude of the blows, as well as the force of the blows and the time period over which the blows are applied.





Oval pads are a good match for cohesive soils and thicker lifts.



Square pads are best suited for semi-cohesive soils and thinner lifts.

The frequency/time relationship accounts for slower working speeds on vibratory compactors. Working speed is important because it dictates how long a particular part of the fill will be compacted. For vibratory compactors, a speed of 1-2,5 km/h (0.6-1.6 mph) for rock and clay, and 2-5 km/h (1.2-3 mph) for gravel and sand will provide the best results.

Smooth drum vibratory compactors were the first vibratory machines introduced. They are most effective on granular materials with particle sizes ranging from large rocks to fine sand. They are also used on semi-cohesive soils with up to 50 percent cohesive soil content. Lift thicknesses vary according to the size of the compactor. Whenever large rock is used in the fill, the lifts may be very thick—up to 1,2 m (4 ft) lifts are not unusual. One thing to remember when large rocks are in the fill is that the thickness should be about 300 mm (12 in) more than the maximum rock size. This permits lift consolidation without having large rocks protrude above the surface.

Padfoot drum machines expand the material range to include soils with more than 50 percent cohesive material and a greater percentage of fines. When the pad penetrates the top of the lift, it breaks the natural bonds between the particles of cohesive soil and achieves better compaction results. The pads are involuted to walk out of the lift without fluffing the soil and tapered to help them stay clean. The typical lift thickness for padded drum units on cohesive soil is in the 150-460 mm (6-18 in) range.

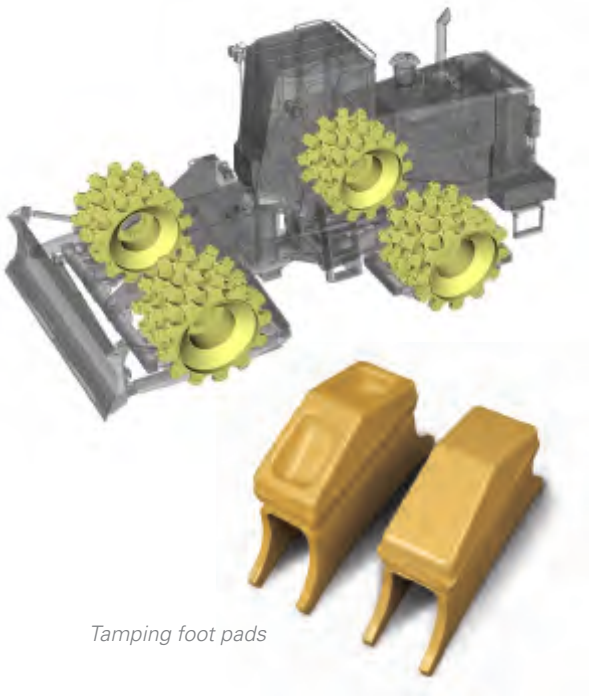
Caterpillar has two pad shapes available: square faced pads and oval-faced pads. Square pads perform well on semi-cohesive soils and thinner lifts of less than 150 mm (6 in). Square pads do a good job of sealing the surface.

Oval pads have less surface area than square pads, so they apply greater ground pressure than square pads. This allows the pad to penetrate deeper into the lift. Oval pads perform better on cohesive soils and thicker lifts of 150-460 mm (6-18 in). Oval pads do not seal the surface as well as square pads.



Caterpillar also offers a padfoot shell kit option for smooth drum compactors. The two-piece shell bolts onto the smooth drum, allowing the compactor to be used on cohesive soils like a normal padfoot drum. Both square pad and oval pad shell kits are available.





Tamping foot pads

Tamping Foot Compactors – Tamping foot compactors are high-speed, self-propelled, non-vibratory compactors. They usually have four padded steel wheels and are equipped with a dozer blade. Their pads are tapered with a rectangular face.

Tamping foot compactors compact from the bottom of the lift to the top. Because the pads are tapered, they can walk out of the lift without fluffing the soil. Therefore, the top of the lift is also being compacted and the surface is relatively smooth and sealed. Tamping foot compactors are capable of speeds in the 16-32 km/h (10-20 mph) range, but they typically operate in the 10-15 km/h (6-10 mph) range.

Generally, 2-3 cycles (4-6 machine passes) will achieve desired densities in 200-300 mm (8-12 in) lifts although 4 cycles may be needed in poorly graded plastic silt or very fine clay. Tamping foot compactors are effective on all soils except clean sand.

Tamping foot compactors leave a fairly smooth, sealed surface so hauling units are able to maintain a high speed when traveling over the fill. Also, since dozer-equipped tamping compactors do both spreading and compacting, the contractor may be able to reduce the number of track-type spreaders.

Tamping foot compactors are best suited for large projects. They need long, uninterrupted passes to build up speed that generates high production. On lifts greater than 300 mm (12 in) thick, tamping foot compactors are about 2 to 3 times more productive than single drum vibratory compactors. The application, jobsite size and the economics behind decision-making will dictate which kind of machine is best.

Sheepsfoot Compactors – Sheepsfoot compactors derived their name from the fact that early Roman road builders would herd sheep back and forth over base material until the road was compacted. The word "sheepsfoot" became a generic term to describe all types of padded drums. In reality, a sheepsfoot compactor is very different from a padded drum or tamping foot compactor.

A sheepsfoot pad is cylindrical, usually 200 mm (8 in) long. The pad is circular and will range in diameter from 76-127 mm (3-5 in). The pads on tamping foot or padded drums are tapered with an oval or rectangular shape. And the pad face is smaller than the base of the pad—that's an important difference.



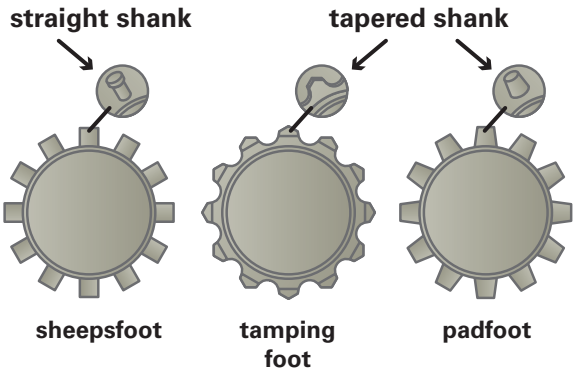
The pads on sheepsfoot drums penetrate through the top lift and actually compact the lift below. When a pad comes out of the soil, it kicks up or fluffs the material. The result is a loose layer of material on top. When more fill is spread, the top lift will be fluffed and the previous layer will be compacted. A sheepsfoot compactor truly compacts from the bottom up.

Using a sheepsfoot compactor has one definite benefit. Because the top lift of soil is always being fluffed, the process helps aerate and dry out wet clays and silts.

But the disadvantages of sheepsfoot compactors are numerous. The loose top-lift material can act as a sponge when it rains and slow the compaction process. The loose material also slows hauling units that deposit fill material, so haul cycle times are increased.

Plus, sheepsfoot compactors can work only at speeds from 6-10 km/h (4-6 mph), which cancels any benefit from impact and vibration. Pressure and manipulation are the only compactive forces exerted on the soil. Usually, 6-10 cycles (12-20 machine passes) are needed to reach target density on 200 mm (8 in) lifts. Sheepsfoot compactors are no longer widely used.

PAD CONFIGURATIONS



Pneumatic Compactors – Pneumatic compactors are used on small-to-medium size soil compaction jobs, primarily on bladed, granular base materials. Often, they are used as a finishing compactor after a vibratory drum compactor completes compaction of the lift. Pneumatic compactors are best suited for sealing the surface, special applications such as compaction of thin lifts, or special requirements dictated by the job.



The compactive forces (pressure and manipulation) generated by the rubber tires work from the top of the lift down to produce density. The amount of compactive force can be varied by altering the tire pressure (the normal method) or by changing the weight of the ballast (done less frequently). The kneading action caused by the staggered tire pattern helps seal and smooth the surface.

Pneumatic compactors can be used on soil and asphalt, an advantage that allows a road-building contractor to use one compactor for multiple stages of construction.





Rear Vibratory Plates – On non-cohesive material, single drum vibratory soil compactors can be equipped with rear vibratory plates, which provide a surface sealing that cannot be achieved with the single drum compactor alone. This enables the operator to account for the gradient of compaction: the single drum vibratory roller achieves deep compaction, and the vibratory plate compacts and seals the surface.

When the vibratory plates are not needed, the plates should be removed from the machine because their weight can reduce the linear load of the drum, which may mean that additional passes will be required to achieve the compaction target.

COMPACTION DEPTH

Assumes density specification is 95% of Standard Proctor and can vary substantially due to different soil conditions.

